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Batch: A1
Division: A

Subject: Programming with Java

ASSIGNMENT No : 3

TITLE : Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

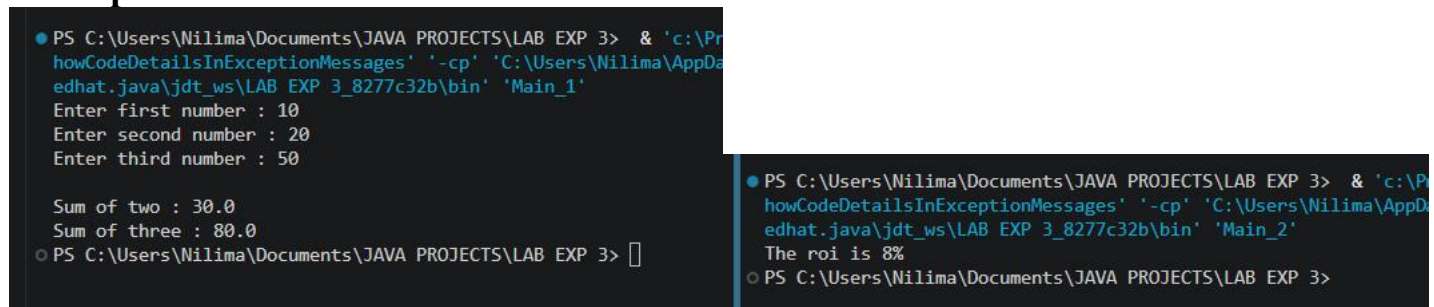
Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

Theory

Method overloading allows multiple methods to have the same name with different parameter lists, enabling compile-time polymorphism. The static keyword is used for variables and methods that belong to the class instead of individual objects. Static members share common data among all objects and reduce memory usage. These concepts improve code flexibility, readability, and efficient resource management.

Output Screenshots



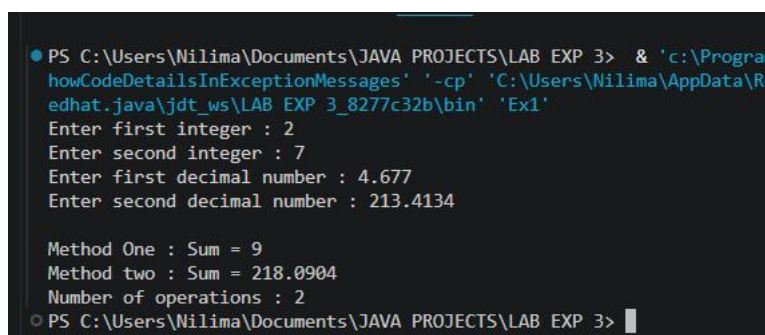
```
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> & 'c:\Program Files\Java\jdk-8.0.60\bin\java.exe' -cp 'C:\Users\Nilima\AppData\Local\Temp\edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Main_1'
Enter first number : 10
Enter second number : 20
Enter third number : 50

Sum of two : 30.0
Sum of three : 80.0
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3>

PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> & 'c:\Program Files\Java\jdk-8.0.60\bin\java.exe' -cp 'C:\Users\Nilima\AppData\Local\Temp\edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Main_2'
The roi is 8%
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3>
```

Exercise

Q1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.



```
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> & 'c:\Program Files\Java\jdk-8.0.60\bin\java.exe' -cp 'C:\Users\Nilima\AppData\Local\Temp\edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Ex1'
Enter first integer : 2
Enter second integer : 7
Enter first decimal number : 4.677
Enter second decimal number : 213.4134

Method One : Sum = 9
Method two : Sum = 218.0904
Number of operations : 2
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3>
```

Q2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

```

PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> java -cp 'C:\Users\Nilima\AppData\Local\Programs\Java\jdk-11.0.10\bin\java.exe;C:\Users\Nilima\AppData\Local\Programs\Java\jdk-11.0.10\bin\edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Ex2'
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 1
Enter the amount : Rs.2000
The total amount is = Rs.2010

----- Billing -----
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 2
Enter the amount : Rs.2000
The total amount is = Rs.2200

----- Billing -----
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 1

PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> java -cp 'C:\Users\Nilima\AppData\Local\Programs\Java\jdk-11.0.10\bin\java.exe;C:\Users\Nilima\AppData\Local\Programs\Java\jdk-11.0.10\bin\edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Ex2'
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 3
Enter the amount : Rs.2000
The total amount is = Rs.2150

----- Billing -----
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 4

Exiting...

----- Total Orders -----
Total orders : 3

PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3>

```

Q3. Method Overriding: A vehicle tracking system needs to calculate the maximum speed of different types of vehicles. A generic vehicle has a base speed. A car has a higher max speed. A bike has an even higher max speed.

```

PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> java -cp 'C:\Users\Nilima\AppData\Local\Programs\Java\jdk-11.0.10\bin\java.exe;C:\Users\Nilima\AppData\Local\Programs\Java\jdk-11.0.10\bin\edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Ex3'
The speed for the vehicle is : 100 km\hr
The speed for the vehicle is : 200 km\hr
The speed for the vehicle is : 300 km\hr

PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3>

```

Q4. Method Overloading: Payment Processing System

A company is building a payment gateway that allows users to pay using different methods:

Credit Card (card number & CVV)

Bank Transfer (account number & IFSC code)

UPI ID

```
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> & 'c:\Pro
howCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Nilima\AppData
edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Ex4'
----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 1
Paid using credit card.

----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 2
Paid using Bank Transfer.

----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 3
Paid using UPI ID.

----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 4

Exiting...
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> |
```

Conclusion

This assignment helped in understanding the concepts of method overloading and static variables and methods in Java. It demonstrated how method overloading enables multiple methods with the same name to perform different tasks based on their parameter lists, improving code flexibility and readability. The use of static variables and methods showed how data and functionality can be shared among all objects of a class, reducing memory usage and simplifying program design. These concepts are fundamental to object-oriented programming and are widely used in developing efficient, reusable, and well-structured Java application.

